

Linear Programming Report

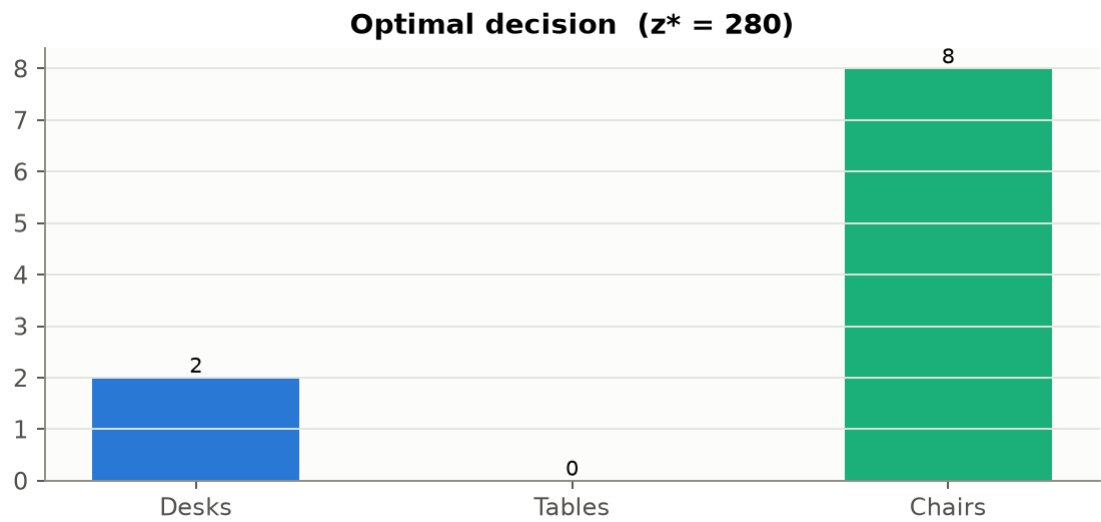
Sensitivity Analysis & Duality

Dakota Furniture

Dakota manufactures desks, tables and chairs using lumber, finishing hours and carpentry hours. Resources are already purchased, so Dakota maximizes total revenue.

Source: Lecture slides 26-38 (dual of a max problem)

Problem type	MAX, 3 variables, 3 constraints
Status	OPTIMAL
Optimal Revenue (\$)	280
Dual status	OPTIMAL
Prepared by	Dr. Islam Ali
Course	Sensitivity Analysis and Duality (Dr. Islam Ali)
Date	28 September 2026, 08:44



1. Model formulation

MAX $z = 60 \text{ Desks} + 30 \text{ Tables} + 20 \text{ Chairs}$
 subject to
 $8 \text{ Desks} + 6 \text{ Tables} + \text{Chairs} \leq 48$ (Lumber)
 $4 \text{ Desks} + 2 \text{ Tables} + 1.5 \text{ Chairs} \leq 20$ (Finishing)
 $2 \text{ Desks} + 1.5 \text{ Tables} + 0.5 \text{ Chairs} \leq 8$ (Carpentry)
 $\text{Desks} \geq 0, \text{ Tables} \geq 0, \text{ Chairs} \geq 0$

Data table

Constraint	Desks	Tables	Chairs	Sense	RHS
Lumber	8	6	1	$\leq$	48
Finishing	4	2	1.5	$\leq$	20
Carpentry	2	1.5	0.5	$\leq$	8
Objective c	60	30	20	max	—

Definitions

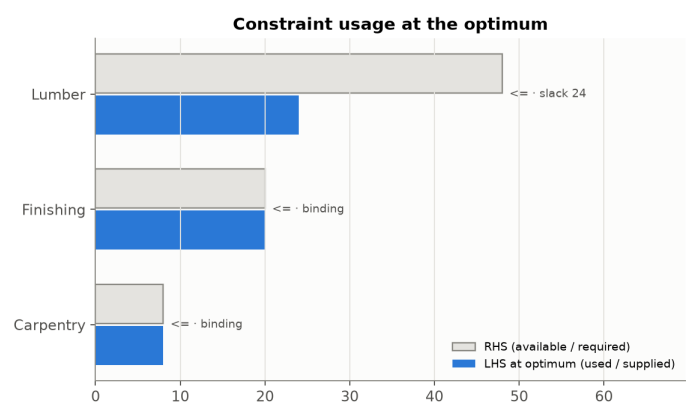
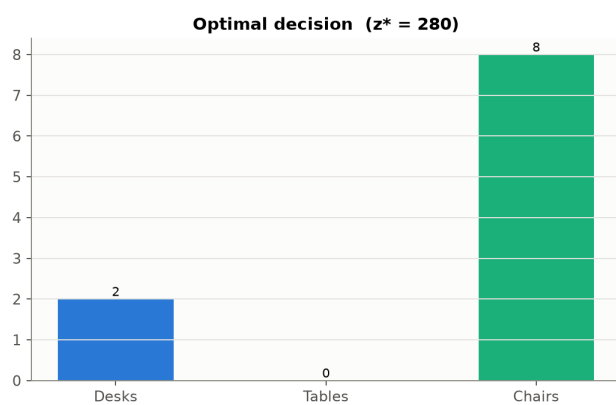
- **Desks:** number of desks produced
- **Tables:** number of tables produced
- **Chairs:** number of chairs produced
- **Lumber:** board feet of lumber
- **Finishing:** finishing hours
- **Carpentry:** carpentry hours

2. Optimal solution

Optimal Revenue (\$): **280**

Variable	Description	Optimal value	Obj. coefficient	Contribution
Desks	number of desks produced	2	60	120
Tables	number of tables produced	0	30	0
Chairs	number of chairs produced	8	20	160

Constraint	LHS	Sense	RHS	Slack/Surplus	Binding
Lumber	24	<=	48	24	no
Finishing	20	<=	20	0	yes
Carpentry	8	<=	8	0	yes



3. Sensitivity analysis

Objective-coefficient ranging (variable cells)

Variable	Value	Reduced cost	Obj. coef	Allow. +	Allow. –	Low	High
Desks	2	0	60	20	4	56	80
Tables	0	-5	30	5	∞	$-\infty$	35
Chairs	8	0	20	2.5	5	15	22.5

Right-hand-side ranging (constraints)

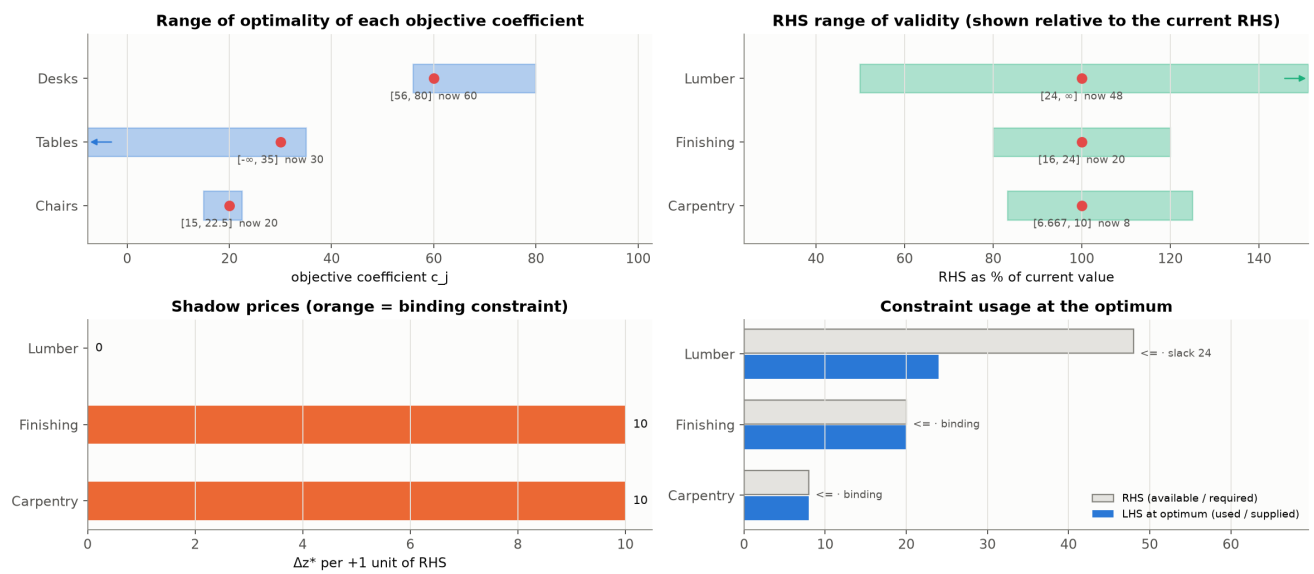
Constraint	LHS	Sense	RHS	Slack	Shadow price	Allow. +	Allow. –	Low	High
Lumber	24	$\leq$	48	24	0	∞	24	24	∞
Finishing	20	$\leq$	20	0	10	4	4	16	24
Carpentry	8	$\leq$	8	0	10	2	1.333	6.667	10

Interpretation

- **Desks** = 2. Its objective coefficient (60) may vary in [56, 80] without changing the optimal solution (one change at a time).
- **Tables** is not used (value 0). Reduced cost -5: its coefficient must increase by about 5 before it becomes attractive.
- **Chairs** = 8. Its objective coefficient (20) may vary in [15, 22.5] without changing the optimal solution (one change at a time).
- **Lumber** is not binding (slack 24), so its shadow price is 0; the RHS can move within [24, ∞] without effect on Revenue (\$).
- **Finishing** is binding. Shadow price 10: each extra unit of RHS changes Revenue (\$) by 10, valid while the RHS stays in [16, 24].
- **Carpentry** is binding. Shadow price 10: each extra unit of RHS changes Revenue (\$) by 10, valid while the RHS stays in [6.667, 10].

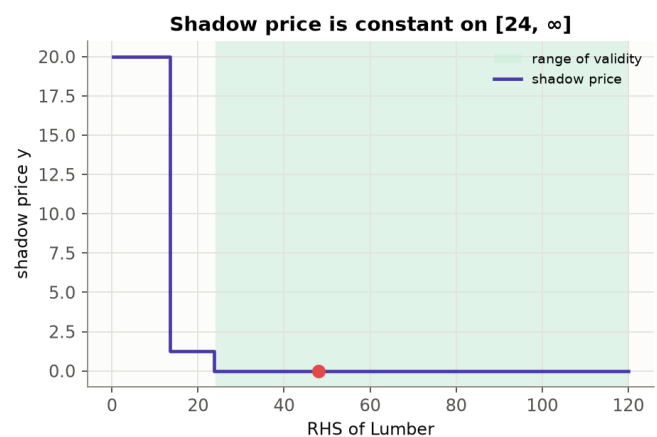
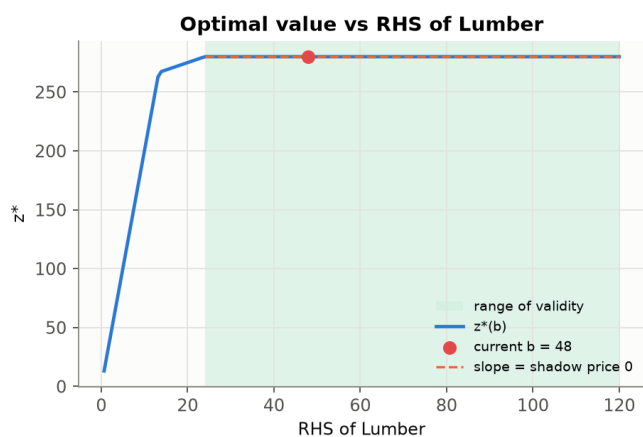
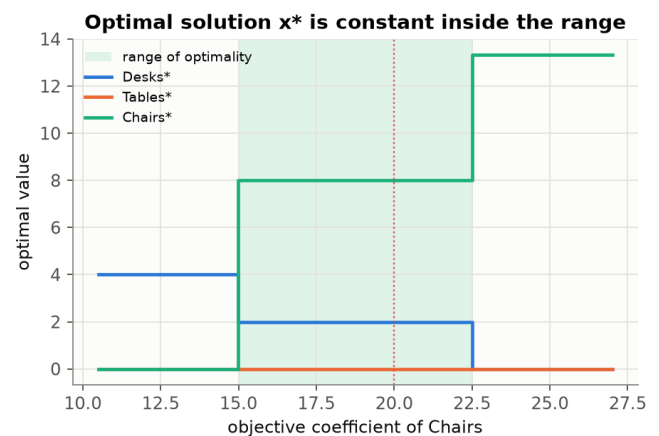
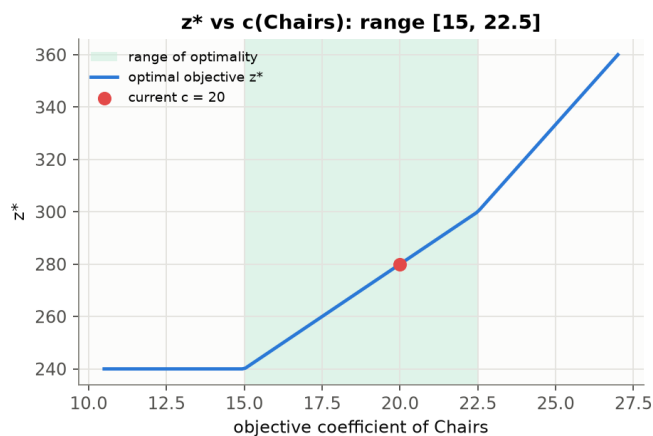
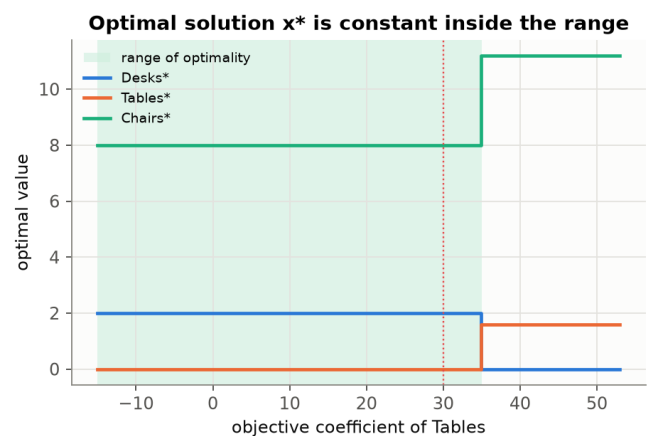
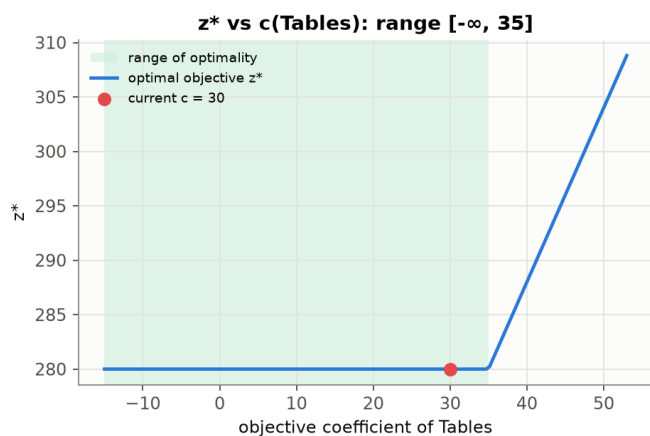
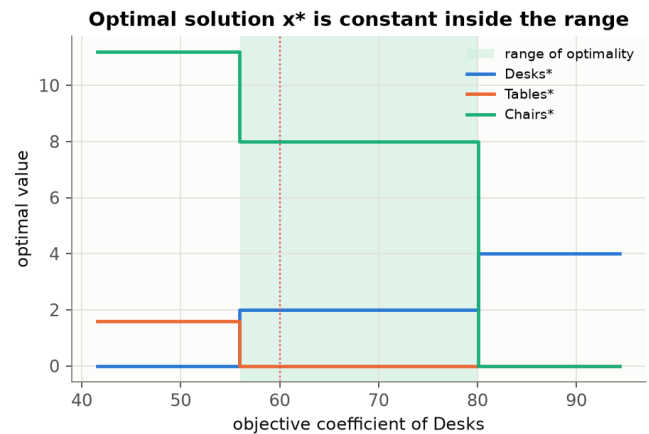
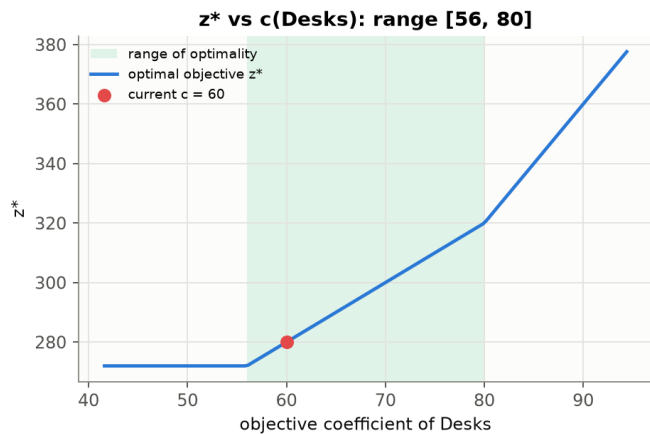
Sensitivity dashboard

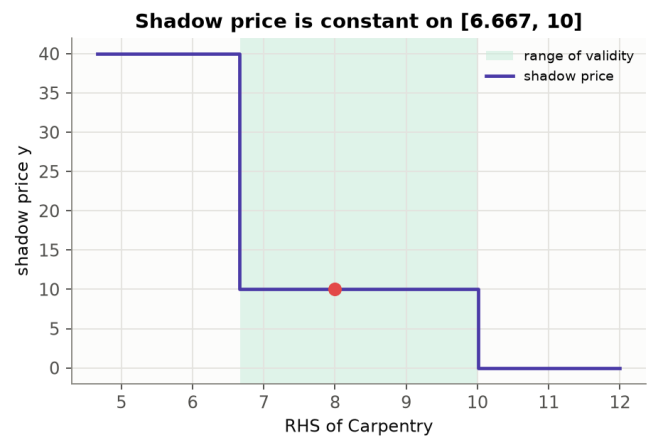
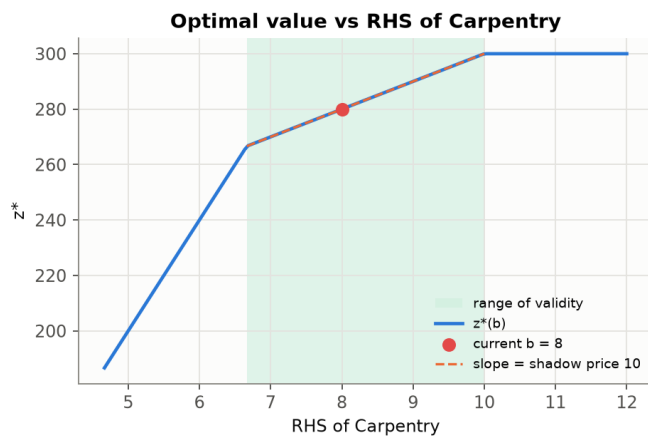
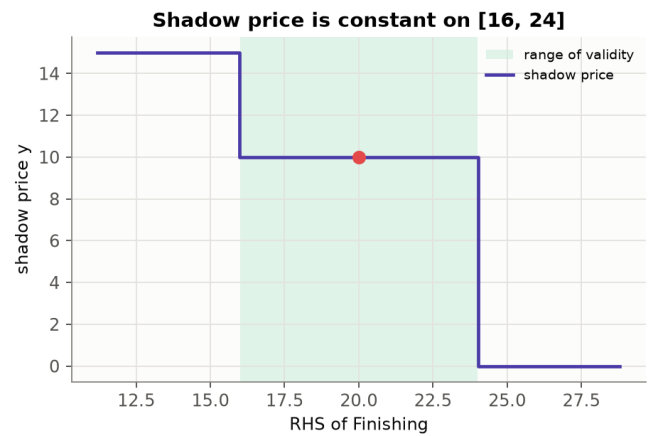
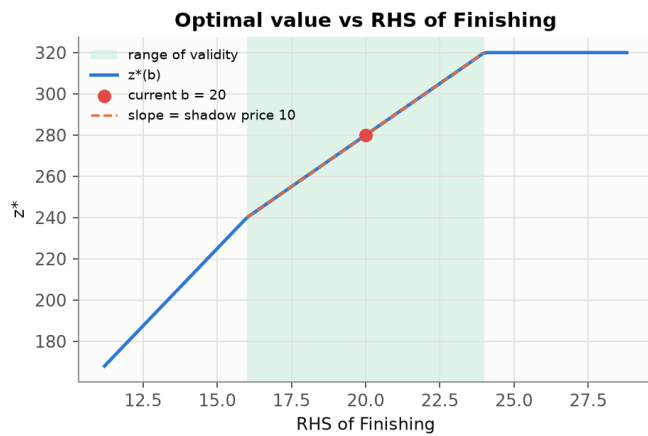
Dakota Furniture: sensitivity dashboard ($z^* = 280$)



4. Parametric analysis

Each chart re-solves the LP for a range of values of one coefficient. The shaded band is the range reported in section 3.





5. What-if scenarios

'Predicted' uses only the sensitivity report (shown when the change is inside its range); 'Actual' re-solves the LP.

Scenario	InRange	Predicted	Actual	NewSolution	Note
c(Tables): 30 → 40	no	–	288	0, 1.6, 11.2	what price makes tables worth producing?
b(Finishing): 20 → 22	yes	300	300	1, 0, 12	two extra finishing hours
b(Lumber): 48 → 40	yes	280	280	2, 0, 8	lose 8 board feet of lumber

6. Duality

Dual formulation

MIN $z = 48 y_{\text{Lumber}} + 20 y_{\text{Finishing}} + 8 y_{\text{Carpentry}}$
 subject to
 $8 y_{\text{Lumber}} + 4 y_{\text{Finishing}} + 2 y_{\text{Carpentry}} \geq 60$ (Desks)
 $6 y_{\text{Lumber}} + 2 y_{\text{Finishing}} + 1.5 y_{\text{Carpentry}} \geq 30$ (Tables)
 $y_{\text{Lumber}} + 1.5 y_{\text{Finishing}} + 0.5 y_{\text{Carpentry}} \geq 20$ (Chairs)
 $y_{\text{Lumber}} \geq 0, y_{\text{Finishing}} \geq 0, y_{\text{Carpentry}} \geq 0$

Dual solution (resource / requirement prices)

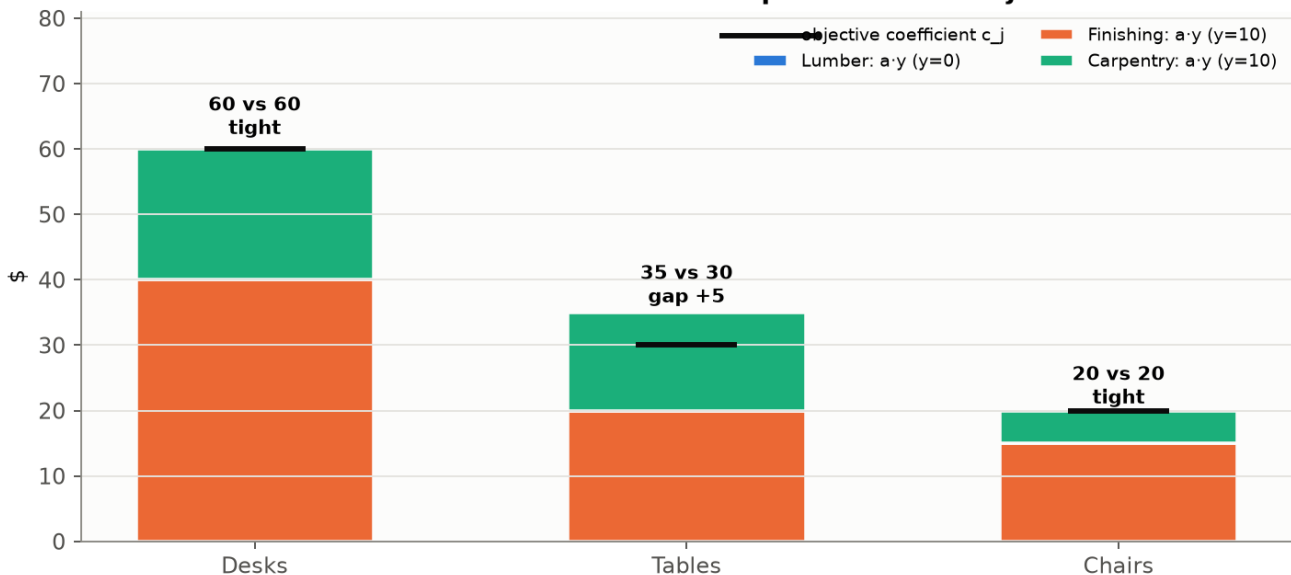
Dual variable	Primal constraint	Value y^*	Primal shadow price	$b_i \cdot y_i$
y_{Lumber}	Lumber	0	0	0
$y_{\text{Finishing}}$	Finishing	10	10	200
$y_{\text{Carpentry}}$	Carpentry	10	10	80

Strong duality check: primal $z^* = 280$, dual $w^* = 280$ (equal ✓). Dual of the dual equals the primal: yes.

Complementary slackness (all products must be 0)

Pair	Slack	DualValue	Product
Lumber slack $\times y_{\text{Lumber}}$	24	0	0
Finishing slack $\times y_{\text{Finishing}}$	0	10	0
Carpentry slack $\times y_{\text{Carpentry}}$	0	10	0
Desks $\times$ surplus of dual constraint Desks	0	2	0
Tables $\times$ surplus of dual constraint Tables	5	0	0
Chairs $\times$ surplus of dual constraint Chairs	0	8	0

Dual constraints: value of resources in each product vs its objective coefficient



Primal (max): 3 constraints × 3 variables

60	30	20	← objective
8	6	1	≤ 48
4	2	1.5	≤ 20
2	1.5	0.5	≤ 8
Desks	Tables	Chairs	

Dual (min): 3 constraints × 3 variables

48	20	8	← objective
8	4	2	≥ 60
6	2	1.5	≥ 30
1	1.5	0.5	≥ 20
y_Lumber	y_Finishing	y_Carpentry	

7. Simplex iterations

Two-phase tableau simplex (entering column = most negative $z_j - c_j$; leaving row = minimum ratio).

Tableau	Method	Phase	Entering	Leaving	Objective	PrimalFeasible	DualFeasible	Point
0	primal	2	Desks	s3	0	yes	no	(0, 0, 0)
1	primal	2	Chairs	s2	240	yes	no	(4, 0, 0)
2	primal	2			280	yes	yes	(2, 0, 8)

Primal simplex - phase 2, tableau 0
enter Desks, leave s3 (min ratio 4)

	Desks	Tables	Chairs	s1	s2	s3	RHS
z	-60	-30	-20	0	0	0	0
s1	8	6	1	1	0	0	48
s2	4	2	1.5	0	1	0	20
s3	2	1.5	0.5	0	0	1	8

Primal simplex - phase 2, tableau 1
enter Chairs, leave s2 (min ratio 8)

	Desks	Tables	Chairs	s1	s2	s3	RHS
z	0	15	-5	0	0	30	240
s1	0	0	-1	1	0	-4	16
s2	0	-1	0.5	0	1	-2	4
Desks	1	0.75	0.25	0	0	0.5	4

Primal simplex - phase 2, tableau 2
optimal tableau

	Desks	Tables	Chairs	s1	s2	s3	RHS
z	0	5	0	0	10	10	280
s1	0	-2	0	1	2	-8	24
Chairs	0	-2	1	0	2	-4	8
Desks	1	1.25	0	0	-0.5	1.5	2

Dual simplex after adding 'MinTables'

New constraint: $0 \text{ Desks} + 1 \text{ Tables} + 0 \text{ Chairs} \geq 1$. marketing demands at least one table Result: **optimal**, new optimum (0.75, 1, 10), Revenue (\$) = 275.

Tableau	Method	Phase	Entering	Leaving	Objective	PrimalFeasible	DualFeasible	Point
0	dual	2	Tables	s_MinTables	280	no	yes	(2, 0, 8)
1	dual	2			275	yes	yes	(0.75, 1, 10)

Dual simplex - phase 2, tableau 0
leave s_MinTables (RHS -1 < 0), enter Tables (min |ratio| 5)

	Desks	Tables	Chairs	s1	s2	s3	s_MinTables	RHS
z	0	5	0	0	10	10	0	280
s1	0	-2	0	1	2	-8	0	24
Chairs	0	-2	1	0	2	-4	0	8
Desks	1	1.25	0	0	-0.5	1.5	0	2
s_MinTables	0	-1	0	0	0	0	1	-1

Dual simplex - phase 2, tableau 1
optimal (primal feasible again)

	Desks	Tables	Chairs	s1	s2	s3	s_MinTables	RHS
z	0	0	0	0	10	10	5	275
s1	0	0	0	1	2	-8	-2	26
Chairs	0	0	1	0	2	-4	-2	10
Desks	1	0	0	0	-0.5	1.5	1.25	0.75
Tables	0	1	0	0	0	0	-1	1